

However, the more common trend is to have camcorders that support both tape and flash memory. Tape is used to store digital video, while Flash is used to store still pictures. The card can then be removed from the camcorder, inserted into a compatible card reader that is in turn connected to a PC or other device (such as a photo-printing printer) for data transfer.

2.2 Recording Formats

2.2.1 DV (Digital Video)

The DV format is the video format standard used by MiniDV tapes. Actually, the DV specification defines both the physical media as well as the DV codec used in recording onto the MiniDV tape. It supports uncomplicated editing, transfer over FireWire, and better quality video than the earlier analogue formats. There are some variants on this standard such as DVCAM and DVCPRO, but these are almost exclusively associated with professional systems.

The raw DV data stream can be transferred over FireWire to a PC using supporting software, such as Windows Movie Maker. A 60-minute MiniDV tape is approximately 13-14 GB in size and takes about an hour to transfer. The raw data stream contains the video and audio data as well as any additional meta-information that can be useful while editing. The files are stored with the .dv and .dif extensions. However, most Windows-based video programs do not support direct DV playback, but require that the it packaged into .AVI containers (files) that only contains the audio and video without the meta information. Exceptions to this are QuickTime Player, VLC Media Player and Mplayer.

2.2.2 MPEG

The MPEG (Moving (or Motion) Pictures Expert Group) standards define various video and audio encoding standards. Depending on the media type, video recording is recorded in one of the three defined standards: MPEG 1, MPEG 2 and MPEG 4. (MPEG 3 was orig-